Whole-Number Division: Up to Four Digits by One-Digit Divisors

three-digit quotients, four-digit dividends, remainders, and estimating to check

Directions. Divide using the long-division cycle: **divide, multiply, subtract, bring down**, and repeat until there are no digits left to bring down. Show all of your work in the space under each problem, then write your final answer on the answer line. If a step leaves a place empty, write a 0 in the quotient — do not skip the place. You can always check a quotient by multiplying it back by the divisor.

1. Divide. $168 \div 4$

Answer: __________________

2. Divide. $375 \div 5$

Answer: __________________

3. Divide. $618 \div 6$ Watch the tens place carefully.

Answer: __________________

4. Nadia wants to estimate $2{,}835 \div 7$ before she divides. She replaces the dividend with the nearby *compatible number* 2,800, because 7 divides it evenly. What estimate does Nadia get?

Answer: ________________

5. Now find the exact quotient. $2{,}835 \div 7$ Then compare it with the estimate you found above — were you close?

Answer: ________________

6. Divide. Write your answer as a quotient with a remainder. $517 \div 4$

Answer: ________________

7. Divide. $891 \div 3$

Answer: ________________

8. A number was divided by 6 and the quotient came out to exactly 143 with nothing left over. What was the number that got divided? Multiplication undoes division — use that.

Answer: ________________

9. Divide. $4{,}872 \div 8$

Answer: ________________

10. A factory has 3,275 pencils. The pencils are packed into boxes, and each full box holds 6 pencils. How many *full* boxes can the factory pack?

Answer: ________________

11. The same factory packs the same 3,275 pencils into full boxes of 6. After every full box is packed, how many pencils are left over?

Answer: ________________

12. A school printed 4,158 raffle tickets and shared them equally among 7 classes. Each class then split its tickets equally among 9 students. How many raffle tickets did each student receive? This takes two divisions. Estimate first so you know about how big each answer should be.

Answer: ________________